

Neural Differential Utility: Accepted Adaptive Service Control and Certified Value-Gradient Learning

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Abstract. Can a provider adapt an externally priced service protocol without reducing customer utility or service? We study persistent tiers, physical stock, costly reconfiguration, and participation at every review. Supporting service prices identify physical decisions under aggregate commitments. For general finite-tree and graph-coupled tier programs, we characterize optimality of an arbitrary feasible outside protocol by continuation prices and bounded switching tensions. Eliminating tensions on a scalar tree gives exact subtree cuts and a full-tree critical-friction program. The characterization includes boundary protocols, multiple commitments, and nonconstant comparators, for which optimal friction values form an interval rather than necessarily an upper ray. The same balance equations yield full-horizon policy certificates. A loss decomposition links value-gradient approximation to accepted decisions: quadratic actor-error bounds require the appropriate acceptance and switching face, while leaving that face can create first-order losses. Independent validation supports a finite-sample certificate for choosing complete learning pipelines. Exact information-value benchmarks are retained. Noncentered, boundary-inclusive experiments measure feasibility repair and actual learning–certification–refinement at matched error tolerances. These results establish structural and implementation guarantees without assuming that learned proposals outperform classical optimization.

Keywords: accepted service control; continuation participation; value-gradient learning.

Subject classifications: Inventory/production: service contracts; Dynamic programming/optimal control: stochastic control; Analysis of algorithms: approximation guarantees.

Area of review: Stochastic Models.

1 Introduction

A service provider can revise both physical capacity and the service tier attached to it. A higher tier earns a premium but can also increase indemnity and maintenance costs; replacing an installed tier consumes resources. Adaptation is useful only when these gains survive the customer’s commitments and comparison with a properly optimized outside protocol. We ask when an accepted adaptive protocol creates value, how switching friction changes that answer, and how a learned implementation can be evaluated under the same acceptance requirements.

We study Neural Differential Utility as joint control of physical activity and a persistent, externally priced contractual state. Regimes, stock, tiers, and realized fulfillment are observable and contractible. The tariff and accounting rule are fixed before optimization. The provider commits to an announced contingent protocol; in the continuation model the customer can leave at every review. There is no private type or hidden effort. The model is an ordinary Markov decision process with expanded state and action spaces, not a claimed escape from stochastic control. Its distinctive comparison preserves the customer’s utility, expected fill, and physical resource commitments relative to an optimized static arrangement.

Our first contribution identifies the structure of accepted adaptation on a full scenario tree. With absolute switching costs, marginal operating rewards must be balanced by continuation prices, boundary scarcity, and bounded switching tensions. On a scalar tree, eliminating the tensions yields cumulative subtree cuts and an exact critical-friction program. This extends the earlier local edge criterion beyond a two-review adjustment. The general balance theorem admits vector tiers, joint smooth graph costs, multiple commitments, signed net-payment rates, boundary solutions, and nonconstant outside protocols. With all temporal and installation edges penalized, a constant protocol with no initial mismatch becomes optimal at sufficiently large friction. For an already-changing outside protocol, the optimality region can instead have an upper endpoint. We give both a complete characterization and a constructive improving policy when the balance fails. A sharp price–capacity margin bounds the amount of accepted value and is exact for the stated quadratic subclass.

Our second contribution connects this structure to value-gradient learning. The same continuation and switching prices enter a full-horizon residual certificate and an exact decomposition of policy loss. That decomposition distinguishes smooth approximation error from costly acceptance slack and switching-face error. On the appropriate exposed face, squared actor-gradient error controls accepted regret. Outside that face, even a small feasible actor error can create a first-order loss. A separate finite-sample result certifies the choice among frozen learning pipelines using independent validation. These results specify what certification means: they concern submitted contingent policies and declared learning/selection procedures, not unrestricted superiority of a neural architecture or arbitrary distribution shift.

Our third contribution supplies reproducible economic and computational comparisons. The exact canonical decomposition retains the accepted values of time, regime information, and inherited tiers under continuous actions and public randomization. Its total improvement over static is 0.63666527; inherited-tier feedback contributes 0.01929038 after time and regime information are available. These exact values illustrate the general theory rather than establish its generality. A new study uses 48 noncentered graph–tree instances, with reoptimized service-specific static tiers and boundary comparators, and 432 deployed pipelines. We compare common scaling, weighted projection, and a continuation-feasible tree-transfer actor;

we then measure actual certification and classical refinement at common tolerances. Earlier adverse transfer results and corrected cached classical timings remain in the companion. No calibrated field application or universal learning speed advantage is inferred from these synthetic experiments.

1.1 Related literature and the incremental contribution

Inventory decisions and service contracts have long been studied jointly. Henig et al. (1997) embed inventory decisions in supply-contract design; Liang and Atkins (2013); Sieke et al. (2012); Katok et al. (2008) study agreements, penalties, coordination, and review periods. Dynamic stocking under service-level agreements is established (Hosseini et al., 2022; Caggiano et al., 2007; Abbasi et al., 2022). Here the static-contract comparator can already adapt physical stock to the regime. The additional decisions are accepted revisions of contractual tiers, not state-aware inventory actions withheld from the benchmark.

Private information and unobserved effort create different constraints. Nasser and Turcic (2019) study temporary contract adjustment with a private demand forecast; Plambeck and Zenios (2003) study incentive-efficient control with unobserved effort. We impose public-state continuation participation, not truth telling or hidden-action incentive compatibility. Combined pricing and inventory optimization is also established (Federgruen and Heching, 1999); our tariff and demand law remain external to the provider’s control.

Constrained Markov decision processes, occupation measures, and Lagrangian resource prices are classical (Altman, 1999). Conditional promised utility supports recursive contracting (Spear and Srivastava, 1987), while self-enforcing wage contracts analyze participation over time (Thomas and Worrall, 1988). Our remaining-payment state is a service-control specialization, not a new principle of dynamic programming. The contribution is the exact continuation-price and switching-capacity characterization, including its cumulative cut form and friction comparative statics, under the stated observable-service institution.

Monotone selections follow Topkis (1978), and hysteretic control has predecessors including Hipp and Holzbaier (1988). Supporting-price stock identification, normal cones, Bregman decompositions, and strong-convexity residual bounds use standard convex tools (Boyd and Vandenberghe, 2004). We state those tools explicitly so that their specialization is not mistaken for independent mathematical novelty. Approximate dynamic programming also uses optimization to connect approximate values and decisions (de Farias and Van Roy, 2003). Our complete-tree audit retains all participation rows and boundary prices rather than relying on a collection of unverified one-step errors.

Derivative-weighted fitting follows Sobolev training (Czarnecki et al., 2017); graph filtering follows Defferrard et al. (2016). Safe and constrained learning studies cumulative constraints and, in some models, unknown safe regions (Satija et al., 2020; Wachi and Sui, 2020). Here the feasible polyhedron is known; learning proposes a policy and a separate calculation certifies it. The acceptance-face theorem and certificate-risk selection result explain the role and limits of this separation. We benchmark SciPy’s sequential quadratic programming implementation alongside the preserved sparse Newton methods (Virtanen et al., 2020); no custom implementation is declared the practical optimization frontier. Forward–backward value-process methods (Ma et al., 1994; Han et al., 2018) and Markov-chain approximation (Kushner and Dupuis, 2001) support the separately preserved continuous-time material, not the finite-tree theorem by analogy.

The main paper contains the institution, essential proofs, canonical comparison, and current implemen-

tation results. Optional predecessor proofs and diagnostic tables are in the electronic companion. The historical supplement preserves the full earlier diffusion, continuous-time, and computational development without making it a prerequisite for understanding the current results.

2 An externally priced inventory contract

There are periods $t = 0, \dots, T - 1$. The observed state is (z, q) , where z is a demand regime in a finite ordered set $\mathcal{Z}$ and $q \in [0, 1]$ is the inherited contract. Conditional demand D_z is nonnegative with finite first moment. A decision consists of stock $S \in \mathcal{S} \subset [0, \bar{S}]$ and contract $\theta \in \Theta \subset [0, 1]$. Both feasible sets are fixed compact chains; finite grids are allowed. Unused stock perishes and unmet demand is lost. Thus no inventory or backlog is carried. The next regime has exogenous transition matrix P , and the next inherited contract equals the chosen contract.

Write the expected overage and shortage as

$$O_z(S) = \mathbb{E}(S - D_z)^+, \quad L_z(S) = \mathbb{E}(D_z - S)^+, \quad C_z(S) = cS + hO_z(S) + pL_z(S). \quad (1)$$

Here $c, h, p \geq 0$ are externally fixed physical-cost coefficients. The provider receives premium $r_z\theta$, pays incremental liability $\kappa\theta L_z(S)$, pays maintenance $m\theta^2$, and pays adjustment $\lambda(\theta - q)^2$, where $\kappa, m, \lambda > 0$. Expected one-period net reward is

$$g(z, q, S, \theta) = r_z\theta - C_z(S) - \kappa\theta L_z(S) - m\theta^2 - \lambda(\theta - q)^2. \quad (2)$$

The expectation in this expression is over demand, not over the decision. Premium rates, liability coefficients, and physical-cost weights cannot be controlled. The contract is a real cash-flow commitment. Its level affects both reward and the future state; it is not an estimator of a preference parameter.

2.1 A customer-accepted master agreement

The contract is a publicly specified menu, not a penalty that the provider can change after performance is observed. Before period zero, a risk-neutral customer and provider sign a master agreement. It specifies the premium schedule $r_z\theta$, the indemnity $\kappa\theta(D_z - S)^+$, the horizon, and a contingent operating protocol. Regime, tier, stock, demand, and fulfillment are observable and contractible. At each review the provider implements the announced protocol after observing (z, q) and before demand is realized. Public randomization is permitted when specified in the agreement. The customer commits to the contingent menu, not to a new purchase at each review. Transfers and compliance with the protocol are enforceable. No hidden effort or private type is assumed.

Let $f_z(S) = \mathbb{E}D_z - L_z(S)$ be expected units served. For a fixed service benefit $\nu > 0$, the customer's incremental one-period utility is

$$u(z, S, \theta) = \nu f_z(S) + \kappa\theta L_z(S) - r_z\theta. \quad (3)$$

Indemnity is a transfer from the provider to this customer. Maintenance and adjustment are real resource expenditures, not payments to the customer. Write $F^\pi = \mathbb{E}^\pi \sum_t \beta^t f_{z_t}(S_t)$, $D = \mathbb{E} \sum_t \beta^t \mathbb{E} D_{z_t}$, and $U^\pi = \mathbb{E}^\pi \sum_t \beta^t u(z_t, S_t, \theta_t)$. Demand is exogenous, so D is independent of the protocol. The agreement is accepted only if

$$U^\pi \geq \underline{U}, \quad F^\pi \geq \gamma D. \quad (4)$$

These are ex ante participation and expected discounted fill commitments. They are not period-by-period chance constraints or a right to renegotiate after each regime realization. A customer with such rights would require continuation participation constraints and an additional promised-utility state.

A fixed retainer A_0 can fund provider participation. Total provider profit is $A_0 + W^\pi$ and total customer surplus is $U^\pi - A_0$; the retainer does not affect policy comparisons. In the numerical study $A_0 = 6$, $\nu = 2$, and the customer's outside alternative is the best fixed-contract arrangement, including the same retainer. Both that arrangement and the accepted adaptive arrangement satisfy a zero provider outside option. We report incremental W and U so that the historical accounting remains exactly comparable.

The institution fixes the menu and designs the contingent protocol subject to acceptance. It does not claim to solve private-information screening or to endogenize every price in the menu. The unconstrained Bellman equation below is the provider relaxation and is also the problem when the acceptance constraints are slack. Section EC.1 solves the binding-acceptance problem. A lower fill rate or customer surplus in the relaxation is not presented as an improvement under the accepted agreement.

For discount $\beta \in (0, 1]$ and zero terminal reward, the Bellman equation is

$$V_t(z, q) = \max_{S, \theta} \left\{ g(z, q, S, \theta) + \beta \sum_{z'} P_{zz'} V_{t+1}(z', \theta) \right\}, \quad V_T = 0. \quad (5)$$

Finite actions make existence immediate. For compact intervals, bounded demand moments, continuity of the stage reward, and induction give a continuous value and an attained maximum. We select the greatest optimal pair when the lattice conditions below hold.

The exact expanded-action representation and the provider-relaxation structure are retained in EC Section EC.4.1. Acceptance is imposed in the following sections.

3 Identification and a unified accepted-control problem

This section supplies a common formulation for the scalar contract model, joint service portfolios, and continuation participation. Critical-fractile analysis and convex first-order optimality are established tools; the supporting-price interpretation identifies their role here. The full-tree cut characterization in Section 4 supplies the switching and continuation structure; the earlier scalar edge criterion is retained in Appendix A.

3.1 Demand primitives and identified physical decisions

Let D_z be nonnegative demand with finite mean, on an exogenous regime process whose occupation probabilities do not depend on the policy. On a compact stock interval $I_z = [l_z, u_z]$, $l_z < u_z$, write

$$C_z(S) = cS + h\mathbb{E}(S - D_z)^+ + p\mathbb{E}(D_z - S)^+, \quad f_z(S) = \mathbb{E} \min(S, D_z), \quad \Psi_z(S) = C_z(S) - \xi f_z(S).$$

Here $\xi > 0$ is a supporting service price, not a contractual transfer chosen by the provider. Denote the demand distribution function by H_z , avoiding confusion between physical fill and resource cost. The critical level is $\tau = (p + \xi - c)/(h + p + \xi)$, with $h + p + \xi > 0$.

Proposition 3.1 (Identified stocks, boundaries, and slack commitments). *Suppose s_z uniquely minimizes Ψ_z on I_z . Then the commitments $F^\pi \geq F^s$ and $C^\pi \leq C^s$ force $S_t = s_{z_t}$ almost surely at every positive-probability review, and both commitments bind. This statement includes randomized policies.*

For an interior minimizer, the one-sided derivatives satisfy

$$\Psi'_{z,-}(S) = c - p - \xi + (h + p + \xi)H_z(S-), \quad \Psi'_{z,+}(S) = c - p - \xi + (h + p + \xi)H_z(S).$$

Thus $H_z(s_{z-}) \leq \tau \leq H_z(s_z)$. At l_z the condition is $\Psi'_{z,+}(l_z) \geq 0$, and at u_z it is $\Psi'_{z,-}(u_z) \leq 0$. A strictly straddling atom or a continuously increasing crossing gives uniqueness; when no crossing lies in the feasible interval the constrained minimizer can instead be an endpoint. The condition $0 < \tau < 1$ alone does not guarantee an interior feasible quantile.

If, separately, $\Psi_z(S) - \Psi_z(s_z) \geq \alpha|S - s_z|^2$ on every I_z , then relaxed commitments imply

$$\mathbb{E}^\pi \sum_t \beta^t |S_t - s_{z_t}|^2 \leq (\delta_C + \xi \delta_F)/\alpha, \quad F^\pi \geq F^s - \delta_F, \quad C^\pi \leq C^s + \delta_C. \quad (6)$$

A sufficient smooth-demand condition is a density at least $\rho > 0$ throughout the entire feasible stock interval, with $\alpha = (h + p + \xi)\rho/2$.

Proof. The displayed derivatives follow by differentiating expected excess and shortage from the left and right. Convexity gives the endpoint and interior conditions. Sum the nonnegative gaps $\Psi_{z_t}(S_t) - \Psi_{z_t}(s_{z_t})$ under the policy's discounted occupation probabilities. Exogeneity makes the comparator use those same probabilities, while acceptance bounds their sum above by $(C^\pi - C^s) - \xi(F^\pi - F^s) \leq 0$. Uniqueness therefore identifies stock at each reachable review. Under quadratic growth the same sum is at least the left side of (6) times α . The density bound gives $\Psi''_z \geq 2\alpha$ almost everywhere; integration and the constrained first-order condition prove the growth inequality even at an endpoint. The smooth-growth conclusion is not attributed to the atomic canonical instance. $\square$

For d stocks with arbitrary correlated demands, a concrete vector extension is $C_z(S) = \sum_j C_{z_j}(S_j)$ and $f_{z_j}(S_j) = \mathbb{E} \min(S_j, D_{z_j})$, with a separate commitment $F_j^\pi \geq F_j^s$ for every service and a single total physical-cost cap. Choose $\xi_j > 0$ and the unique constrained marginal quantiles at levels $(p_j + \xi_j - c_j)/(h_j + p_j + \xi_j)$. On a product stock set, $\Psi_z = C_z - \sum_j \xi_j f_{z_j}$ has the unique vector minimizer. Correlation is immaterial

because these costs are additive expectations. Under coordinate density bounds, (6) holds with a squared Euclidean norm, numerator $\delta_C + \sum_j \xi_j \delta_{F,j}$, and the smallest coordinate curvature. Coupled stock constraints require these coordinate minimizers to be jointly feasible; the statement is not extended to an arbitrary vector service functional by assertion.

3.2 Joint resources and explicit information restrictions

On a finite exogenous tree, let $x_n \in [0, 1]^d$ be the tier vector at decision node n , $w_n > 0$ its discounted probability, and $a_n \in \mathbb{R}^d$ its net payment rates. The fixed initial tier is q_0 . Identified physical cost C^s is constant. Set

$$\mathcal{R}(x) = \sum_n w_n \{M_n(x_n) + A_n(x_n - x_{\text{pa}(n)})\} + \mathcal{G}(x), \quad (7)$$

$$\begin{aligned} W(x) &= p^\top x - \mathcal{R}(x) - C^s, & p_n &= w_n a_n, \\ X &= \{x \in [0, 1]^{Nd} : Bx \leq b, Ex = e\}. \end{aligned} \quad (8)$$

All resource functions are differentiable and convex on a neighborhood of the box. The joint function $\mathcal{G}$ includes, for example, $\sum_n w_n \zeta_n \sum_{(i,j) \in \mathcal{E}_n} v_{nij} (x_{ni} - x_{nj})^2$. This is genuinely nonseparable within a portfolio. It is not represented by a sum of univariate functions. The equations $Ex = e$ identify actions sharing the same permitted information; equality multipliers remain unrestricted, rather than being disguised as payment prices. Each continuation row of B uses discounted subtree payments. Multiplication by a positive node probability converts a conditional participation inequality to this unconditional form.

Let $\bar{x}$ embed the reoptimized static comparator, with commitments defined from that comparator and $\bar{x} \in X$. Put $g = p - \nabla \mathcal{R}(\bar{x})$. Define

$$T_X(\bar{x}) = \overline{\{s(y - \bar{x}) : s \geq 0, y \in X\}}, \quad N_X(\bar{x}) = \{v : v^\top (y - \bar{x}) \leq 0 \forall y \in X\}.$$

Lemma 3.2 (Standard first-order alternative specialized to acceptance). *An accepted policy strictly improves $W(\bar{x})$ if and only if $g \notin N_X(\bar{x})$. Equivalently, the following explicit linear program has a positive value:*

$$\max_d g^\top d \quad \text{s.t.} \quad B_i d \leq 0 \ (B_i \bar{x} = b_i), \quad Ed = 0, \quad \|d\|_\infty \leq 1, \quad d_j \geq 0 \ (\bar{x}_j = 0), \quad d_j \leq 0 \ (\bar{x}_j = 1).$$

For such a direction, let $s_{\max} = \sup\{s \geq 0 : \bar{x} + ud \in X \text{ for } 0 \leq u \leq s\}$. If the directional second derivative of $\mathcal{R}$ is at most L_d on this segment, then

$$W(\bar{x} + sd) - W(\bar{x}) \geq sg^\top d - L_d s^2 / 2. \quad (9)$$

For $L_d > 0$, take $s = \min\{s_{\max}, g^\top d / L_d\}$.

Proof. Concavity gives $W(y) - W(\bar{x}) \leq g^\top (y - \bar{x})$. Conversely every direction in the displayed polyhedral tangent cone is feasible on a short segment, and a positive derivative gives improvement. Tangent–normal polarity gives the equivalence. Twice integrating the directional curvature bound proves (9). This is the

ordinary first-order test and curvature estimate for convex optimization, not a new optimality principle (Boyd and Vandenberghe, 2004). $\square$

Lemma 3.3 (Resource loss and acceptance slack). *At an optimizer x^* , suppose $\eta \geq 0$, $\xi \in \mathbb{R}^{\text{rows}(E)}$, and $v \in N_{[0,1]^{Nd}}(x^*)$ satisfy $p - \nabla \mathcal{R}(x^*) = B^\top \eta + E^\top \xi + v$ and $\eta^\top (b - Bx^*) = 0$. For every $x \in X$,*

$$W^* - W(x) = D_{\mathcal{R}}(x, x^*) + \eta^\top (b - Bx) + v^\top (x^* - x),$$

where $D_{\mathcal{R}}(x, x^*) = \mathcal{R}(x) - \mathcal{R}(x^*) - \nabla \mathcal{R}(x^*)^\top (x - x^*)$. Every term is nonnegative.

Proof. Add and subtract the linearization of $\mathcal{R}$, substitute stationarity, and use complementary slackness and $E(x - x^*) = 0$. The signs follow from convexity, feasibility, and the outward-normal definition. For a differentiable convex objective over a nonempty polyhedron, first-order optimality and the polyhedral normal-cone representation provide these multipliers; no interior Slater point is required for that representation. This standard KKT decomposition (Boyd and Vandenberghe, 2004) separates real resource inefficiency from unused customer commitments in the present model. It is not claimed as a novel algebraic identity. $\square$

4 Full-tree switching, continuation prices, and critical friction

The scalar edge test in Appendix A becomes a capacitated balance condition when changing a tier has an absolute cost. The balance formulation also accommodates nonconstant outside protocols, boundary tiers, several commitments, and vector services. It does not divide by a tariff and therefore remains meaningful for zero or negative net payments. Negative net payments can arise when expected indemnity exceeds the premium; they need not be interpreted as negative gross prices.

4.1 The general balance condition

Let $P = \{x \in [l, u] : Ax \leq b, Ex = e\}$ be a nonempty compact polyhedron of contingent service tiers. The rows of A can include several discounted commitments at each history; the rows of E include information restrictions or an exact root payment commitment. Fix a feasible comparison protocol $z \in P$. Neither constancy nor interiority of z is assumed. All probabilities remain exogenous. Write

$$W_\lambda(x) = r(x) - \lambda T(x), \quad r(x) = p^\top x - \mathcal{R}_0(x), \quad T(x) = \sum_{a \in \mathcal{A}} \kappa_a |(Dx - h)_a|, \quad \lambda \geq 0. \quad (10)$$

Here $\mathcal{R}_0$ is convex and differentiable on a neighborhood of the box, $\kappa_a > 0$, and D is a specified incidence matrix. A temporal row is the difference between a tier and its parent's tier. An initial-installation row has one nonzero entry and its installed tier in h . Spatial switching or coordination edges may be included. Smooth graph coupling remains in $\mathcal{R}_0$. Omitted switching edges simply have no row. Let $I(z) = \{i : A_i z = b_i\}$, $g = \nabla r(z)$, and let $N_{[l, u]}(z)$ denote the outward box normal: its j th component is nonpositive at a lower boundary, nonnegative at an upper boundary, and zero in the interior. Fixed coordinates permit either sign.

Theorem 4.1 (Capacitated continuation balance). *The protocol z maximizes (10) on P if and only if there exist continuation prices $\mu_i \geq 0$ for $i \in I(z)$, unrestricted equality prices ν , a box normal $\xi \in N_{[l,u]}(z)$, and edge tensions s satisfying*

$$g = A_{I(z)}^\top \mu + E^\top \nu + \xi + D^\top s, \quad (11)$$

$$s_a = \lambda \kappa_a \operatorname{sign}(Dz - h)_a \quad \text{if } (Dz - h)_a \neq 0, \quad (12)$$

$$|s_a| \leq \lambda \kappa_a \quad \text{if } (Dz - h)_a = 0. \quad (13)$$

Consequently the set of friction coefficients for which this fixed protocol is optimal is a closed interval in $[0, \infty)$, possibly empty, a singleton, or unbounded. Its endpoints are the infimum and supremum of λ in the displayed linear program. These are linear programs in primitive derivatives, not the original nonlinear control problem.

Proof. The convex minimization form has objective $\mathcal{R}_0(x) - p^\top x + \lambda T(x)$. It is finite and continuous on a neighborhood of P . The subdifferential sum rule and the polyhedral normal formula give optimality exactly when $g \in \partial(\lambda T)(z) + N_P(z)$. The subgradients of each absolute value give (12)–(13); the polyhedral normal is $A_{I(z)}^\top \mu + E^\top \nu + \xi$. No strictly feasible point is needed for this normal representation. Conversely these equations and the supporting inequalities for $\mathcal{R}_0$, T , and P show $W_\lambda(x) \leq W_\lambda(z)$ for every feasible x . Finally, all conditions are linear in $(\lambda, s, \mu, \nu, \xi)$, including the signs of box normals. Their projection onto the scalar λ is a polyhedron, hence a closed interval with the stated possibilities. An unbounded endpoint is allowed. $\square$

The subgradient principle is standard convex analysis (Boyd and Vandenberghe, 2004); matrix absolute-value penalties and their dual treatment also appear in the generalized lasso (Tibshirani and Taylor, 2011). The economic content here is the separation of marginal operating reward into continuation prices, boundary scarcity, and bounded switching tensions. On a tree, the tensions can be eliminated entirely to obtain explicit subtree cuts.

4.2 Subtree cuts and an isotonic critical-friction program

For a scalar tree let $c_n > 0$ be discounted net-payment coefficients. Suppose the budget rows are $\sum_{j \geq n} c_j x_j \leq b_n$ for nonroot nodes and the root has exact payment equality. Let z be any feasible outside protocol. In this subsection D contains the temporal edges and initial installation: $h_o = q_0$, $h_n = 0$ off the root. Set $z_{\text{pa}(o)} = q_0$. Introduce a node price ρ such that

$$\rho_o \in \mathbb{R}, \quad \rho_n - \rho_{\text{pa}(n)} \begin{cases} \geq 0, & n \in I(z), \\ = 0, & n \notin I(z), \end{cases} \quad (n \neq o). \quad (14)$$

Here $I(z)$ indexes binding continuation budgets only. Define the cumulative residual on a subtree by

$$H_n(\rho, \xi) = \sum_{j \geq n} (g_j - c_j \rho_j - \xi_j). \quad (15)$$

Corollary 4.2 (Exact cut characterization). *The comparison protocol z is optimal if and only if some ρ satisfying (14) and some $\xi \in N_{[l,u]}(z)$ satisfy, at every node including the root,*

$$\begin{cases} H_n(\rho, \xi) = \lambda \kappa_n \text{sign}(z_n - z_{\text{pa}(n)}), & z_n \neq z_{\text{pa}(n)}, \\ |H_n(\rho, \xi)| \leq \lambda \kappa_n, & z_n = z_{\text{pa}(n)}. \end{cases} \quad (16)$$

For weak root acceptance replace the free root price by $\rho_o \geq 0$ when that budget binds, and by $\rho_o = 0$ when it is slack. If initial installation has no absolute cost, use $H_o = 0$ instead of the root capacity condition.

If $z_n = q_0 = \bar{\theta} \in (l_n, u_n)$ for every node, all continuation caps equal the payments of z , and all switching weights including the root are positive, the exact critical friction is finite and equals

$$\lambda_c = \min_{\rho: \rho_n \geq \rho_{\text{pa}(n)}} \max_n \frac{|\sum_{j \succeq n} (g_j - c_j \rho_j)|}{\kappa_n}. \quad (17)$$

The static protocol is optimal exactly for $\lambda \geq \lambda_c$. The root price in (17) is unrestricted under root equality.

Proof. The sum of the root multiplier and all binding-budget multipliers on a node's ancestral path is precisely ρ_n . This is equivalent to (14). For the rooted incidence matrix, $D^\top s = g - c\rho - \xi$ reads $s_n - \sum_{v: \text{pa}(v)=n} s_v = g_n - c_n \rho_n - \xi_n$. Summing over a subtree cancels all interior edges and leaves $s_n = H_n$. Substitution into Theorem 4.1 proves both directions, including installation and the weak-root variant. In the constant interior case there are no box normals or saturated edges. Minimizing the remaining capacities gives (17). Taking a constant ρ , for example zero, gives a finite feasible upper bound; the linear program attains its finite minimum. Increasing λ relaxes every remaining capacity, proving the ray characterization. $\square$

This program has a linear number of node variables and local balance constraints when subtree sums are represented recursively. Evaluating all cuts for given prices takes one backward tree pass. We do not assert that solving the linear program has linear running time. With $\lambda = 0$, the balance reduces to $g_n/c_n = \rho_n$, recovering the scalar edge criterion. For a two-review chain, eliminating its single payment-neutral transfer recovers the threshold in Corollary A.2. In contrast to that special case, (17) allows simultaneous transfers across all branches. For vector services, signed coefficients, several budgets, or extra graph edges, Theorem 4.1, rather than division by c_n , is the applicable formulation.

4.3 Constructive improvements and comparative statics

Let $K = T_P(z)$ be the polyhedral tangent cone. Set $v = Dz - h$ and

$$\phi_z(d) = \sum_{a: v_a \neq 0} \kappa_a \text{sign}(v_a)(Dd)_a + \sum_{a: v_a = 0} \kappa_a |(Dd)_a|.$$

Thus ϕ_z is the directional derivative of switching cost, including boundary installation. The following alternative gives an implementable improving policy when the capacity test fails.

Proposition 4.3 (Certified direction and friction sensitivity). *The balance test fails exactly when the piecewise-linear program*

$$\delta = \max_{d \in K, \|d\|_\infty \leq 1} \{g^\top d - \lambda \phi_z(d)\} \quad (18)$$

has positive value. Absolute-value epigraphs turn it into a linear program. For a maximizing direction d , choose $a_{\max} > 0$ so that $z + ad \in P$ and no nonzero component of $Dz - h$ changes sign for $0 \leq a \leq a_{\max}$. If the smooth resource curvature along this segment is at most L_d , then

$$W_\lambda(z + ad) - W_\lambda(z) \geq a\delta - L_d a^2/2. \quad (19)$$

When $T(z) = 0$, the accepted gain $G(\lambda) = \max_{x \in P} W_\lambda(x) - r(z)$ is nonnegative, nonincreasing, and convex in λ . If x_λ is an optimizer, $-T(x_\lambda)$ is a subgradient of G ; optimal total switching is nonincreasing with friction. Tightening any commitments while keeping z feasible cannot increase either G or its critical friction. Increasing edge weights cannot increase the critical friction of a zero-switching comparator.

Proof. For a convex objective on a polyhedron, nonnegative directional derivatives in every tangent direction are equivalent to global minimality. Negating this statement gives (18); homogeneity permits the bounded normalization. A tangent direction is feasible for a sufficiently short segment. Bounds from inactive inequalities, box faces, and the finitely many nonzero v_a give the required positive $a_{\max}$. On that segment $T(z + ad) - T(z) = a\phi_z(d)$ exactly. Integrating the smooth curvature bound proves (19); take $a = \min\{a_{\max}, \delta/L_d\}$ if $L_d > 0$. For $T(z) = 0$, G is a supremum of affine nonincreasing functions of λ , and z supplies the zero lower bound. The maximizing affine function supplies its subgradient. Adding the two optimality inequalities at two friction values orders their optimal switching costs. Feasible-set inclusion proves the commitment statement. Larger edge weights reduce every nonstatic reward without changing $r(z)$, proving the final statement. $\square$

For a zero-switching comparator the equivalent variational threshold is the positive part of $\sup_{d \in K, T'(z;d) > 0} g^\top d / T'(z;d)$. A direction with $T'(z;d) = 0$ and $g^\top d > 0$ makes this threshold infinite. The initial edge with positive weight eliminates such directions on a scalar rooted tree. This convention matters when installation or an entire component is unpenalized.

For a nonconstant comparison protocol, it would be incorrect to extrapolate the monotone-friction conclusion. The relative gain is still convex, but its supporting slope is $T(z) - T(x_\lambda)$, which can have either sign. An exact example fixes a root tier at zero and permits a child tier $x \in [0, 1]$, with $r(x) = x - (x - 1)^2/2$ and $T(x) = x$. The outside protocol $z = 1$ is optimal exactly on $0 \leq \lambda \leq 1$. For $1 < \lambda < 2$ the optimizer is $2 - \lambda$, and for $\lambda \geq 2$ it is zero. Increasing friction therefore makes this already-changing outside protocol improvable. The interval assertion of Theorem 4.1, rather than a universal lower threshold, covers this case.

4.4 A sharp price–capacity margin for the value of adaptation

The balance test also yields a quantitative measure of accepted value. This measure optimizes the certificate’s prices, rather than choosing arbitrary multipliers and interpreting a possibly loose bound. For a positive

definite metric H , define

$$C_\lambda(z) = N_P(z) + \partial(\lambda T)(z), \quad \Delta_\lambda = \min_{c \in C_\lambda(z)} \|g - c\|_{H^{-1}}. \quad (20)$$

This is a convex quadratic program in the same continuation prices and edge tensions as the linear balance test. It measures the marginal reward that cannot be absorbed by commitments, boundaries, and switching capacities.

Theorem 4.4 (Sharp price–capacity margin). *Suppose $\mathcal{R}_0$ is m_0 -strongly convex and L_0 -smooth in metric H , with $0 < m_0 \leq L_0$. Let c^* attain (20) and put $d = H^{-1}(g - c^*)$. Then $d \in T_P(z)$ and*

$$g^\top d - \lambda \phi_z(d) = \|d\|_H^2 = \Delta_\lambda^2. \quad (21)$$

For any $a_{\max} > 0$ such that the segment $z + [0, a_{\max}]d$ is feasible and preserves every nonzero switching sign, write $a = \min\{a_{\max}, 1/L_0\}$. The exact accepted gain satisfies

$$(a - L_0 a^2/2) \Delta_\lambda^2 \leq \max_{x \in P} W_\lambda(x) - W_\lambda(z) \leq \frac{\Delta_\lambda^2}{2m_0}. \quad (22)$$

If $a_{\max} \geq 1/L_0$, the lower bound is $\Delta_\lambda^2/(2L_0)$. If the smooth resource is a quadratic with Hessian $m_0 H$ and $a_{\max} \geq 1/m_0$, both bounds coincide and $z + d/m_0$ is globally optimal. Thus the same price–capacity program gives both the threshold and the exact gain on this quadratic subclass.

Proof. The set $C_\lambda(z)$ is a nonempty closed polyhedron, so the metric projection exists. Write $c^* = n^* + t^*$ with $n^* \in N_P(z)$ and $t^* \in \partial(\lambda T)(z)$. The projection variational inequality gives $d^\top(c - c^*) \leq 0$ for all $c \in C_\lambda(z)$. Varying only the normal component and taking it equal to zero or $2n^*$ gives $d^\top n^* = 0$; varying it arbitrarily gives $d \in N_P(z)^\circ = T_P(z)$. Varying only the switching component gives $d^\top t^* = \sup_{t \in \partial(\lambda T)(z)} d^\top t = \lambda \phi_z(d)$. Since $(g - c^*)^\top d = \|d\|_H^2$, this proves (21).

For any feasible x , the normal inequality, the switching supporting inequality, and strong convexity bound its gain by $(g - c^*)^\top(x - z) - m_0\|x - z\|_H^2/2$. Completing the square gives the upper bound. On the specified segment switching cost has its directional affine form. Smoothness and (21) give the lower bound at $z + ad$. When the smooth resource is the stated quadratic and $a = 1/m_0$, the lower bound reaches the global upper bound. This also proves the claimed policy’s global optimality. $\square$

For zero-switching comparators, the critical friction is the smallest λ for which $\Delta_\lambda = 0$. A small nonzero margin with limited box clearance can correspond to a much smaller realizable gain; (22) keeps that clearance explicit. With a nonconstant comparator, sign clearance also matters. These qualifications prevent interpreting an unconstrained quadratic distance as the accepted gain when a proposed step crosses a tier boundary or changes a pre-existing switching sign.

Budget values and normalization. For fixed dynamics, objective, and equality constraints, the optimal value is concave and nondecreasing as a function of feasible upper-budget vectors b . Mixing two feasible policies proves concavity; an optimal nonnegative budget multiplier is a supergradient by weak duality. These

are sensitivity statements for externally fixed tariffs, not endogenous tariff-design results. In the recursion (40), b and $\bar{b}_n$ are *conditional* remaining discounted payments, measured from node n . Multiplication by $w_n = \beta^{t(n)} \Pr(n)$ gives the unconditional subtree budget used in A . The child term uses βP_{nv} , not the unconditional child probability a second time. Several commitments add a vector budget state and one such inequality per commitment. Resource and promise states are classical constructions (Altman, 1999; Spear and Srivastava, 1987; Thomas and Worrall, 1988); the present contribution is the continuation-price and switching-capacity structure they induce in this observable service-control problem.

5 What accepted adaptivity buys

The occupation program establishes a feasible improvement, but an information decomposition must impose its commitments in every class. We now solve that comparison with continuous tiers, rather than transferring an unconstrained information increment to the accepted problem. A supporting-price argument first identifies the physical policy from the commitments themselves.

5.1 The service and resource commitments identify stock

Lemma 5.1 (Physical-policy identification). *Let s_z be a fixed-stock rule and suppose that, for some $\xi > 0$,*

$$C_z(S) - C_z(s_z) - \xi\{f_z(S) - f_z(s_z)\} \geq 0 \quad (23)$$

for every feasible z, S , with equality only at $S = s_z$. Demand regimes are exogenous. Any protocol with $F^\pi \geq F^s$ and $C^\pi \leq C^s$ must select $S_t = s_{z_t}$ almost surely at every positive-probability review. Both commitments then hold with equality. For the canonical instance, $s_z = z + 2$ and $\xi = 1$ satisfy (23), for both the original stock menu and every stock in $[0, 6]$.

Proof. The expected discounted sum of the left side of (23) is nonnegative. Feasibility bounds that sum above by $(C^\pi - C^s) - \xi(F^\pi - F^s) \leq 0$. Every nonnegative summand at a reachable state must therefore vanish. Strictness identifies stock, and exogeneity gives equality of both totals. In the instance, $C_z(z+2) = 1.03+0.3z$ and $f_z(z+2) = 1.5+z$. On the half-unit menu, the smallest positive supporting gap is exactly $9/40$. The functions are piecewise linear between the demand atoms, all of which belong to that menu. The same unique minimizer consequently holds on the entire stock interval. The positive gap $9/40$ is a menu gap, not a uniform gap on continuous stocks approaching $z+2$. $\square$

This is not unconstrained stock elimination inside a constrained problem. Both the service floor and the physical-cost cap are used to prove that no other stock decision is feasible. Under looser commitments, we retain all stock choices in the occupation program.

Set $w_{tz} = \beta^t \Pr(z_t = z)$, $a_z = r_z - \kappa L_z(s_z)$, and $A = \sum_{t,z} w_{tz} a_z$. At the canonical stocks, $a_z = 0.36+1.4z$. Let $\bar{\theta}$ be the optimized continuous static tier, and put $\bar{P} = A\bar{\theta}$. The remaining acceptance condition is a payment budget,

$$P^\pi := \mathbb{E}^\pi \sum_t \beta^t a_{z_t} \theta_t \leq \bar{P}, \quad U^\pi = \nu F^s - P^\pi. \quad (24)$$

The accepted reward is $W^\pi = P^\pi - C^s - \mathbb{E}^\pi \sum_t \beta^t [m\theta_t^2 + \lambda(\theta_t - q_t)^2]$. Thus matching the customer's utility fixes expected net payments. The remaining economic benefit must be a reduction in maintenance and reconfiguration resources, not an unreported transfer from the customer.

5.2 Randomization and the accepted information hierarchy

We define randomization before comparing information. An ex ante public random tape R , independent of demand regimes, may select a constant tier, a time schedule, or a map of time and current regime. These classes have rules $f(R)$, $f_t(R)$, and $f_t(z, R)$, respectively. They include deterministic rules and public mixtures of them. They do not silently grant a regime-only rule access to earlier regimes through an unrecorded memory state. Full protocols may use the entire observed history; their optimum will admit a Markov implementation.

Proposition 5.2 (Derandomization on a continuous accepted menu). *Under Lemma 5.1, continuous tier feasibility $[0, 1]$, and convex maintenance and adjustment costs, public randomization does not improve the accepted static, time-only, or time-regime value. An inherited-only randomized rule with deterministic initial tier has the same accepted value as a time-only rule. For the canonical quadratic problem, a deterministic full-state rule also attains the unrestricted accepted optimum.*

Proof. Replace each restricted rule by its expectation over R , holding the regime path fixed. The resulting action remains in $[0, 1]$ and in the same information class. Expected payments do not change. Jensen's inequality decreases each maintenance cost and each adjustment cost, including the jointly randomized difference of consecutive tiers. The fixed physical policy preserves the other commitments. Without regime inputs, an inherited-only rule generates a random time schedule from q_0 and R ; the preceding argument applies. For full histories the conditional-mean construction need not preserve the displayed Markov parametrization. The deterministic Markov rule and a matching unrestricted Lagrangian upper bound constructed below establish attainment directly. None of these statements asserts derandomization on a coarse, nonconvex tier menu. $\square$

For a multiplier $\eta \geq 0$, maximize $W^\pi - \eta P^\pi$ and add $\eta \bar{P}$. This is a different sign convention from the customer-utility multiplier in (EC.5), with the constant customer-service term removed. Write its full-state value as

$$J_t^\eta(z, q) = -A_t q^2 + B_{tz} q + D_{tz}, \quad A_T = B_{Tz} = D_{Tz} = 0. \quad (25)$$

When the maximizers are interior, backward completion of squares gives

$$\begin{aligned} K_t &= m + \lambda + \beta A_{t+1}, & L_{tz} &= (1 - \eta) a_z + \beta \sum_{z'} P_{zz'} B_{t+1, z'}, \\ A_t &= \lambda - \lambda^2 / K_t, & B_{tz} &= \lambda L_{tz} / K_t, \\ D_{tz} &= -C_z(s_z) + \beta \sum_{z'} P_{zz'} D_{t+1, z'} + L_{tz}^2 / (4K_t), & \theta_t^\eta(z, q) &= \lambda q / K_t + L_{tz} / (2K_t). \end{aligned} \quad (26)$$

Endpoint verification is essential: the affine rule must lie in $[0, 1]$ for every $q \in [0, 1]$, not just along one simulated path. We verify both endpoints in rational arithmetic for every period and regime in the canonical certificate.

The restricted classes require their own optimization. Give each reachable time–information pair one tier variable x_i . Their rewards have the form

$$W(x) = -x^\top Hx + b^\top x - c_0, \quad c^\top x \leq \bar{P}, \quad H \succ 0. \quad (27)$$

Here the maintenance terms contribute mw_i to the diagonal; an adjustment edge from variable j to variable i contributes its discounted transition weight times $\lambda(x_i - x_j)^2$. The initial installation cost contributes $\lambda(x_0 - q_0)^2$. There are one, eight, and twenty-two variables in the static, time, and time–regime problems. For an interior solution,

$$x = \frac{1}{2}H^{-1}(b - \eta c), \quad \eta = \max \left\{ 0, \frac{c^\top H^{-1}b - 2\bar{P}}{c^\top H^{-1}c} \right\}.$$

The matrix is not a full-state Bellman relaxation of the regime restriction: a single variable is shared by every history with the same permitted information.

Theorem 5.3 (Exact accepted information values). *For the canonical primitives, the commitments of the optimized continuous static contract, and continuous stock and tier intervals, the accepted values in Table 1 are exact rational numbers. Each value is attained without public randomization. The accepted increment from observing the inherited tier, after time and current regime have already been admitted, is 0.019290378765 The total gain over the optimized static contract is 0.636665271611*

Proof. The exhaustive static stock-envelope calculation yields the stated $\bar{\theta}$ and stocks. Lemma 5.1 then reduces every feasible protocol, not just the proposed ones, to (24). For each restricted class, the displayed solution of (27) has every tier strictly between zero and one. Rational arithmetic verifies its stationarity equation, nonnegative multiplier, primal feasibility, and complementary slackness. Strict concavity proves global optimality in the deterministic class; Proposition 5.2 includes its public mixtures.

In the full problem, A_t is independent of η and B_{tz} is proportional to $1 - \eta$. Forward first moments make P^{θ^η} affine in η . Solving $P^{\theta^\eta} = \bar{P}$ gives $\eta = 0.180271243733$ Rational endpoint checks verify interior maximization in every Bellman step. The affine policy is feasible, and its Lagrangian bound is equal to its reward. Forward second moments evaluate maintenance and adjustment exactly. This bounds every randomized history-dependent protocol from above and attains the bound with a deterministic Markov policy. The files record the rational values, multipliers, and policy coefficients, rather than only the decimals in the table. $\square$

The inherited-tier increment is approximately three percent of the total accepted gain. Most of the gain is due to current-regime reallocation; attributing the entire improvement to contractual memory would be incorrect. Nevertheless, inherited information has strictly positive value under the same customer commitments, not only in a provider relaxation. The continuous optimum also exceeds the certified 0.1-

menu optimum by 0.011524959639.... The earlier lottery is therefore not the source of the economic improvement.

5.3 A primitive mechanism for a robust improvement

Theorem 5.4 (A customer-neutral adaptive direction). *Consider a feasible constant tier $\bar{\theta} \in (0, 1)$ with fixed stocks s_z , linear net payments $a_z\theta$, and twice continuously differentiable maintenance M near $\bar{\theta}$, with $M'(\bar{\theta}) > 0$. Suppose that at a review $t \geq 1$ two regimes have positive discounted weights w_L, w_H and $0 < a_L < a_H$. For every finite quadratic adjustment coefficient, a sufficiently small deterministic time-regime perturbation strictly improves provider reward while preserving customer utility, fill, and physical cost exactly. The conclusion holds in a neighborhood of primitives preserving these strict inequalities.*

Proof. Change only that review's tiers by εd_z , where

$$d_H = (w_H a_H)^{-1}, \quad d_L = -(w_L a_L)^{-1}, \quad d_z = 0 \text{ otherwise.}$$

Then $\sum_z w_z a_z d_z = 0$ and $\sum_z w_z d_z = 1/a_H - 1/a_L < 0$. Payments and physical decisions are unchanged in expectation. The first-order reward gain is $\varepsilon M'(\bar{\theta})(1/a_L - 1/a_H) > 0$. All changed adjustment costs are second order because the baseline tier is constant at this and the next review; the initial installation is unaffected. If $|M''| \leq L_M$ locally, the absolute quadratic cost is at most

$$\varepsilon^2 \{L_M/2 + \lambda + \beta \lambda \mathbf{1}_{\{t < T-1\}}\} \left(\frac{1}{w_H a_H^2} + \frac{1}{w_L a_L^2} \right).$$

Choose ε small enough to keep both tiers interior and make this term smaller than the positive linear gain. Continuity preserves a strictly positive margin in a sufficiently small neighborhood. For $M(\theta) = m\theta^2$, the displayed first- and second-order terms give the exact payoff difference, not just an asymptotic expansion. $\square$

This argument does not require translated demand, a coarse menu, or a lottery. It does require positive and heterogeneous net payment rates and an interior static tier with increasing maintenance cost. It proves existence of an improvement, not that every parameter perturbation has a large gain. It also explains the role of the customer constraint: concentrating a fixed expected payment budget on higher-payment regimes permits a lower average maintained tier. Reconfiguration is worthwhile when those savings exceed its resource cost.

5.4 Allowing the customer to leave at each review

The ex ante institution can be weakened without introducing hidden action or changing the tariff. At every observed regime history h at review t , let the customer's outside option be the remaining service utility of the same static contract. The provider remains committed to the announced protocol, but the customer may leave at each such review. Physical stock is still identified by the service and cost commitments. Continuation

participation therefore requires

$$\mathbb{E} \left[\sum_{s=t}^{T-1} \beta^{s-t} a_{z_s} \theta_s \middle| h \right] \leq \bar{\theta} \mathbb{E} \left[\sum_{s=t}^{T-1} \beta^{s-t} a_{z_s} \middle| h \right] \quad \text{for every reachable } h. \quad (28)$$

These constraints cannot be replaced by the root participation inequality. On the finite regime tree they are linear, while maintenance and adjustment give a strictly concave quadratic objective. The canonical tree has 3,280 decision nodes. Its policy may depend on history; a promise-state representation is an alternative, not an assumption that the original three-coordinate Markov policy remains sufficient.

We solve this convex program numerically, round its deterministic tiers to rational numbers, and contract the vector slightly toward zero to enforce every continuation inequality exactly. The resulting reward is $-5.417032209823 \dots$, an improvement of $0.453078763224 \dots$ over the static contract. A rational feasible-policy and weighted-residual certificate places the interim-participation optimum less than 6.95×10^{-7} above this reward. Thus the exit option costs approximately 0.18358651 relative to full customer commitment, but does not eliminate the gain. EC Section EC.3 gives the upper-bound proof and the complete policy audit. No compensating side payment or change in the fixed menu is used to obtain this result.

6 Learning on acceptance faces and certified deployment

The structural balance condition also provides a common language for learned and classical implementations. This section distinguishes three questions: whether a candidate is feasible, how far its reward can be from optimal, and whether learning selects candidates with useful out-of-sample guarantees. None of these questions is answered by a solver's success flag.

6.1 The same balance equations certify a complete contingent policy

Suppose $\mathcal{R}_0$ is m_0 -strongly convex in a positive definite metric H , that is, its supporting quadratic has term $m_0 \|y - x\|_H^2 / 2$. A numerical candidate x must first belong to P . Choose any $\mu \geq 0$, any equality price v , nonnegative upper and lower bound prices u^+, u^- , and edge tensions s satisfying (12)–(13) at x , so that $D^\top s \in \partial(\lambda T)(x)$. Define

$$\begin{aligned} e_x &= \nabla r(x) - D^\top s - A^\top \mu - E^\top v - u^+ + u^-, \\ C(x) &= \frac{\|e_x\|_{H^{-1}}^2}{2m_0} + \mu^\top (b - Ax) + (u^+)^\top (u - x) + (u^-)^\top (x - l). \end{aligned} \quad (29)$$

The vector u without a superscript denotes the upper tier bound, not a multiplier.

Theorem 6.1 (Balance-residual policy certificate). *Every quantity in (29) is nonnegative except possibly individual residual coordinates, and*

$$0 \leq \max_{y \in P} W_\lambda(y) - W_\lambda(x) \leq C(x).$$

For polynomial smooth resources and rational primitives, candidate, prices, and tensions, feasibility and this bound are exactly checkable when H^{-1} and m_0 are rational. This certificate covers boundary policies and absolute switching. Its zero-residual, zero-complementarity condition is exactly Theorem 4.1 at the candidate.

Proof. Strong concavity of r and the supporting inequality for λT bound $W_\lambda(y) - W_\lambda(x)$ above by $(\nabla r(x) - D^\top s)^\top (y - x) - m_0 \|y - x\|_H^2/2$. Substitute the residual. Feasibility bounds the budget contribution by $\mu^\top (b - Ax)$ and the box contributions by the two displayed slacks. The equality contribution vanishes. Completing the square bounds the remaining residual term by $\|e_x\|_{H^{-1}}^2/(2m_0)$. All inputs of the resulting polynomial/rational expression can be replayed without trusting the candidate generator. $\square$

This is a specialization of the standard strong-convexity bound, not a claim that residual inequalities were invented here. Unlike a stand-alone unconstrained gradient norm, it retains the continuation and boundary prices that also determine accepted adaptivity. In the quartic experiment $H = \text{diag}(w)$ and $m_0 = 2m$, recovering the earlier certificate and adding explicit box prices.

For a feasible comparison protocol z , let $\hat{x}$ be whichever of x, z has larger exactly evaluated reward. A valid gap for the selected decision is

$$\min\{W_\lambda(x) + C(x), W_\lambda(z) + C(z)\} - W_\lambda(\hat{x}). \quad (30)$$

The use of both upper bounds is optional but can be substantially tighter than carrying the failed proposal's bound through the gate. A prescribed tolerance is met only when the selected decision's bound meets it. Otherwise the deployed algorithm must refine, return a separately certified alternative, or explicitly report that the tolerance was not attained.

6.2 Why the acceptance face matters for gradient learning

Consider a fixed feasible set and define the forcing-value function $J(p) = \max_{x \in P} \{p^\top x - \mathcal{R}_0(x) - \lambda T(x)\}$. Strong convexity gives a unique optimizer $x^*(p)$ and $\nabla J(p) = x^*(p)$. For the weighted forcing used in the experiments, $\nabla_p J = \text{diag}(w)x^*$ instead. A scalar potential therefore produces compatible value and policy estimates, but compatibility alone does not make the estimated actor feasible.

Let μ^*, ν^*, ξ^*, s^* be optimal prices and tensions at x^* . Define the convex exposed set

$$\begin{aligned} \mathcal{F}_* = \{y \in P : & \mu^{*\top} (b - Ay) = 0, \quad \xi^{*\top} (x^* - y) = 0, \\ & \lambda T(y) - \lambda T(x^*) - s^{*\top} D(y - x^*) = 0\}. \end{aligned} \quad (31)$$

The last equality selects the face of the piecewise-linear switching cost supported by s^* . It is automatically satisfied when $\lambda = 0$.

Theorem 6.2 (Gradient accuracy, acceptance prices, and policy loss). *For every feasible y ,*

$$\begin{aligned} J(p) - [p^\top y - \mathcal{R}_0(y) - \lambda T(y)] &= D_{\mathcal{R}_0}(y, x^*) + \mu^{*\top} (b - Ay) + \xi^{*\top} (x^* - y) \\ &\quad + \lambda T(y) - \lambda T(x^*) - s^{*\top} D(y - x^*). \end{aligned} \quad (32)$$

All four terms are nonnegative. Suppose additionally that $\mathcal{R}_0$ has L_0 -Lipschitz gradient in metric H on P . For any predicted actor a , its metric projection $y = \Pi_{\mathcal{F}_}^H(a)$ satisfies*

$$0 \leq J(p) - W_\lambda(y) \leq \frac{L_0}{2} \|a - x^*\|_H^2. \quad (33)$$

In particular, for a scalar compatible critic use $a = \nabla \widehat{J}(p)$ (or the appropriately weight-normalized gradient). For any distribution on problems with uniformly bounded L_0 , the same inequality holds after expectation whenever these face conditions hold problem by problem.

Proof. Add and subtract the smooth linearization at x^* and use $p - \nabla \mathcal{R}_0(x^*) = A^\top \mu^* + E^\top \nu^* + \xi^* + D^\top s^*$. Complementary slackness and $E(y - x^*) = 0$ give (32). Convexity, feasibility, the outward normal, and the switching subgradient give the four signs. On $\mathcal{F}_*$ only the smooth Bregman term remains, bounded by $L_0 \|y - x^*\|_H^2 / 2$. Projection onto a closed convex set containing x^* is nonexpansive relative to x^* , so $\|y - x^*\|_H \leq \|a - x^*\|_H$. Taking expectations proves the final assertion. $\square$

This theorem is an approximation-to-decision result, not an assertion that the unknown optimal face is available for free. Verified optimal prices identify it; an unverified predicted face does not establish (33). The residual certificate remains valid without knowing it. The decomposition explains why a common scaling repair can lose reward even when actor distance is small: it can introduce first-order slack in expensive continuation commitments. Indeed, on $[0, 1]$, maximizing $x - (x - 1)^2 / 2$ gives $x^* = 1$, while the feasible actor $1 - \epsilon$ loses $\epsilon + \epsilon^2 / 2$, not merely a quadratic amount. Ignoring the boundary-price term would yield a false gradient-to-regret theorem. The full-face projection in this example is the singleton $\{1\}$.

6.3 A finite-sample certificate for selecting a learning pipeline

The preceding results do not certify arbitrary distribution shift. They can, however, turn independent validation into a precise learning statement. Suppose M frozen candidate-generation pipelines, including all repair, fallback, and refinement rules, are fixed before a validation sample is drawn. The pipelines may have been fitted on separate training data. On problem I , pipeline j returns a feasible policy with regret bounded by a measurable certificate $C_j(I) \in [0, B]$. A uniform range bound B must be established from the problem class, not obtained by discarding large observed bounds. When a separate valid range bound on regret is available, taking its minimum with a residual certificate is legitimate.

Proposition 6.3 (Certificate-risk selection). *For n independent validation problems drawn from the same*

distribution as a future problem, with probability at least $1 - \delta$ simultaneously for all M pipelines,

$$\mathbb{E} \text{Regret}_j(I) \leq \frac{1}{n} \sum_{i=1}^n C_j(I_i) + B \sqrt{\frac{\log(M/\delta)}{2n}}. \quad (34)$$

The statement also holds for the pipeline selected by minimizing the empirical certificate mean. Moreover its expected certificate is at most the best pipeline's expected certificate plus $2B\sqrt{\log(2M/\delta)/(2n)}$ with probability at least $1 - \delta$.

Proof. Condition on the training data and hence on the frozen pipelines. Apply the one-sided bounded-variable Hoeffding inequality to each certificate mean and take a union bound. Expected regret is no larger than expected certificate. Simultaneity permits validation-based selection. For the final assertion use the two-sided inequality, then compare the selected empirical mean to that of a population-minimizing pipeline and add the two deviations. Unconditioning preserves both probability statements. $\square$

The concentration argument is standard (Hoeffding, 1963); the certified object is the selected *complete decision pipeline*, not a neural architecture or an unrestricted distribution-shift claim. A validation set used to change features or tolerances no longer qualifies as independent validation for the displayed guarantee. The new computational study below is a crossed stress test, not a numerical claim that this bound is tight. Constrained-policy learning and value-based safety constructions study different sources of uncertainty (Satija et al., 2020; Wachi and Sui, 2020); here the feasible polyhedron is known and deterministic certification is separated from candidate quality.

7 Noncentered contracts and matched-accuracy implementation

We test the implementation consequences of the balance and acceptance-face results, not only whether a floating-point optimizer terminates. The current experiment complements, rather than replaces, the exact canonical economic comparison and the earlier nonlinear transfer study. Every adverse predecessor outcome remains in EC Section EC.7.

7.1 Design and reoptimized comparators

The smooth resource objective is quartic maintenance plus quadratic temporal and spatial coupling,

$$\mathcal{R}_0(x) = \sum_n w_n \left\{ m \|x_n\|^2 + \gamma \sum_j x_{nj}^4 + \lambda_s \|x_n - x_{\text{pa}(n)}\|^2 + \zeta \sum_{(i,j) \in \mathcal{E}} \omega_{ij} (x_{ni} - x_{nj})^2 \right\}.$$

Here $m = 3/2$, $\lambda_s = 3/5$, discount is $9/10$, and the initial installed tier is one half. The subscript on λ_s distinguishes smooth adjustment from the absolute-switching parameter of Section 4. The current computational study sets that absolute-switching term to zero; the full nonsmooth theorem is instead checked by independently formulated linear programs and exact examples. Net customer tariffs are $a_{nj} = 1 + (j \bmod 3)/4$. Provider reward uses forcing $p = a + r$, where r is an externally specified linear resource

saving. A fixed resource offset of 100 per service–review can make resource accounting nonnegative on the generated box; it cancels from all reported gains and derivatives. This experiment does not endogenize prices or change the demand occupation law.

Four fixed graph families contain twelve independently generated instances each: three-service, three-review cycles; four-service, four-review random-edge graphs; six-service, four-review geometric graphs; and eight-service, four-review weighted-block graphs. Thus contingent plans have 21, 60, 90, and 120 coordinates. Parameters, graph generators, and random seeds are recorded before candidate evaluation. Replication identifiers are reused across families, forming common-random-number blocks; these are resampled jointly, not treated as 48 independent draws from one population. The latter two families have quartic curvature two and three and spatial coupling one half and one, respectively. Forcing is generated from heterogeneous rational service levels plus uncentered node shocks; it is not normalized to make one half optimal.

For each instance we optimize a separate constant-in-time tier for every service, including initial installation and spatial coupling. This vector static class is stronger than the single common-tier benchmark used in the original centered study. We round the optimizer to rational tiers and independently bound its static-class loss. The largest unnormalized static gap is 5.09×10^{-14} . Every subtree cap is then defined from that recorded outside protocol, without reoptimizing the cap for the adaptive method. Fifteen of the 48 outside protocols have at least one boundary tier, and the recorded tiers range from zero to one. Reported adaptive gains are relative to the rational outside protocol; its tiny static approximation bound is disclosed rather than described as exact equality with an irrational optimizer.

The first four previously frozen R10 training seeds supply both scalar-compatible critics, with no retraining on the new instances. They use the same 63 softplus features and the same scalar observations as before. The known static value and actor are re-anchored to the new comparator, but the fitted readout coefficients do not change. This is a deliberate transfer test. It is neither a claim that these four seeds exhaust training variability nor a test of an oracle that was trained on the shifted problems. The old “value-only” fit means no derivative targets in its loss; its shared feature scaling uses derivative geometry and is not derivative-free preprocessing.

7.2 Repair, gate, and actual refinement

We record raw box-clipped actors, all pre-repair continuation violations, the global scaling factor, and the exact reward effect of repair. A weighted Euclidean projection solves a different problem from radial scaling: it finds the closest feasible actor in the discounted tier metric, not the highest-reward actor. A separate tree-flow actor uses nonnegative child-to-parent payment transfers. Each transfer increases a parent tier by its flow divided by the parent’s discounted tariff and decreases the child’s tier by the flow divided by the child’s discounted tariff. Every nonroot subtree payment therefore changes by the negative incoming flow, and root payment is unchanged. A common box-limited step preserves feasibility. These three approaches expose different parts of the acceptance geometry; none is assumed to identify the optimal exposed face for free.

The deployed learned pipeline generates an actor, projects, rounds and exactly repairs it, audits its reward and residual, and gates against the static protocol. It then actually runs SLSQP refinement when

necessary. The cold classical pipeline starts from the same static protocol. Both use the identical quartic objective, analytic gradients, box bounds, every subtree constraint, and the same rational certificate with explicit upper- and lower-bound prices. The common audit schedule checks the initial candidate, every fifth iteration, and termination. Each pipeline stops only when the exact rational upper gap, divided by discounted service–reviews, is at most the prescribed tolerance. A numerical success flag cannot satisfy this stopping rule.

Table 2 gives two tolerances and charges the learned pipeline for candidate generation, projection, audit, gating, and measured refinement. Comparator construction and common setup are recorded separately. All 432 new deployments meet the tighter tolerance, and their exactly evaluated final reward is at least the static reward. The maximum normalized final gap is 9.74×10^{-8} . An independent verifier imports neither the optimizer nor numerical linear-algebra libraries; across current and predecessor re-audits it checks 2682 candidate certificates, 35742 subtree inequalities, and 535658 tier inequalities. Five timing repetitions after a warmup, with randomized pipeline order, are reported separately in EC Table 6; they are not treated as additional economic observations.

These matched-accuracy runs do not show a learned speed advantage over cold SLSQP. They establish an executable fallback that meets a stated accepted-value tolerance, including when transfer fails. A learned proposal’s low generation time is not used to conceal the cost of obtaining the final certificate. We also execute the cached classical fallback on all forty original 992-coordinate weighted-block learned proposals. Its largest normalized final gap is 4.97×10^{-10} ; the median measured generation–audit–fallback time is 423.71 milliseconds. This is an explicit classical fallback, not a claim that those failed actors provide useful Newton warm starts.

7.3 What the transfer evidence identifies

EC Table 4 reports every raw and repaired outcome. Of the 384 new learned proposals, 162 remain worse than static after weighted projection. Such failures are not omitted or relabeled as improvements. Projection can avoid some unnecessary scaling loss, but cannot generally correct a severely inaccurate critic. The tree-flow parameterization enforces the direction of continuation-compatible payment transfers; it does not replace optimization of their economic magnitudes. This agrees with Theorem 6.2: feasibility and closeness alone do not establish the optimal-face condition or eliminate all first-order price losses.

The mean paired projection-gain difference between derivative-weighted and value-only actors is 0.043611 per service–review. A descriptive crossed bootstrap, resampling training seeds and common replication blocks jointly across the four fixed graph families, has percentile interval $[-0.404368, 0.450732]$. Its width and inclusion of zero preclude a reproducible superiority conclusion. This interval addresses different variation from R10’s training-seed interval conditional on eight fixed problems. Neither is an inference over an unspecified population of graph families. The finite-sample selection theorem requires separately frozen pipelines, an independent validation distribution, and a valid uniform bound; it is not retroactively asserted by this stress test.

On the original weighted-block family, learned residual gaps are over a thousand times the classical gain for typical proposals. Using the static upper bound tightens the gate’s error estimate but still leaves a large

gap. EC Table 5 reports both ratios. Thus a no-harm gate is a feasibility/economic safeguard, not a substitute for accuracy refinement. The new endpoint-oracle study also varies the finite-difference step and certified value error; EC Table 7 shows why derivative targets require an explicit error budget. The synthetic evidence supports audited implementation and diagnoses failure mechanisms, rather than supplying a field calibration or a universal architecture ranking.

8 Conclusions and operational implications

Accepted adaptation is a reallocation of real resources under commitments, not a transfer taken from the customer. On a tree, cumulative continuation prices and switching capacities determine which reallocations survive the exit option. The full-tree criterion extends the scalar edge test and distinguishes a zero-switching comparator's critical friction from the interval of frictions supporting an already-changing outside protocol. Boundary and multiple-budget cases belong to the same balance system.

Learning enters through a precise approximation-to-decision connection. The acceptance-face decomposition identifies both the quadratic benefit of accurate gradients and the first-order losses that generic feasibility repair can introduce. Residual and validation certificates provide distinct deterministic and statistical guarantees. Neither substitutes for measuring the complete deployment pipeline. The new comparisons retain severe raw transfer failures, show how actual refinement attains declared accuracy, and separate those findings from an unsupported claim of neural speed or gradient-supervision superiority. Calibrating a service operation, endogenizing tariffs, and treating private information remain distinct extensions; they are not silently assumed in the present observable-state results.

A Primitive structure under continuation participation

Continuation participation changes which customer-neutral reallocations are permissible. We give a complete scalar tree criterion and a quantitative improvement that can be checked from local primitives, without solving the defining nonlinear control program. The graph formulation above remains available for portfolios; the scalar result is not silently promoted to a graph theorem.

A.1 An edge test rather than a high-dimensional direction search

Let o be the root of a finite decision tree. Assume $a_n > 0$, $w_n > 0$, scalar continuous tiers, and an interior constant comparator $\bar{\theta}$. Define $c_n = w_n a_n$ and the accepted, payment-neutral class

$$\sum_n c_n (x_n - \bar{\theta}) = 0, \quad \sum_{j \succeq n} c_j (x_j - \bar{\theta}) \leq 0 \quad (n \neq o), \quad 0 \leq x_n \leq 1. \quad (35)$$

Here $j \succeq n$ includes n itself. Root equality holds customer utility exactly fixed; every nonroot inequality prevents the remaining expected payment from exceeding the static outside protocol. Identified fill and physical cost stay fixed. This is a subclass of weak acceptance, so every strict improvement obtained here also improves the weakly accepted problem.

Let $\mathcal{R}$ be convex and differentiable, and define the primitive marginal reward per payment unit at the comparator by

$$k_n = \frac{w_n a_n - \partial_n \mathcal{R}(\bar{\theta} \mathbf{1})}{w_n a_n}. \quad (36)$$

For scalar maintenance and centered smooth switching, $A'_n(0) = 0$ and $\nabla \mathcal{G}(\bar{\theta} \mathbf{1}) = 0$, this is $k_n = 1 - M'_n(\bar{\theta})/a_n$ at every nonroot node. At the root, subtract also $A'_o(\bar{\theta} - q_0)/a_o$. Thus the test permits nonquadratic maintenance, arbitrary finite smooth friction, and nonstationary tariffs. Initial installation is not omitted.

Theorem A.1 (Complete continuation-compatible edge criterion). *In (35), the static protocol is globally optimal if and only if*

$$k_n \geq k_{\text{pa}(n)} \quad \text{on every nonroot edge.} \quad (37)$$

If some edge $u \rightarrow v$ has $\Delta = k_u - k_v > 0$, set $d_u = 1/c_u$, $d_v = -1/c_v$, and all other components to zero. Then $\bar{\theta} \mathbf{1} + s d$ is feasible for

$$0 \leq s \leq s_{\max} = \min\{c_u(1 - \bar{\theta}), c_v \bar{\theta}\},$$

and strictly improves the reward for all sufficiently small positive s . If $d^\top \nabla^2 \mathcal{R}(\bar{\theta} \mathbf{1} + s d) d \leq L_d$, its gain is at least $s \Delta - L_d s^2/2$. The edge test costs $O(N)$ comparisons once the primitive marginal costs have been evaluated.

Proof. For any feasible x , put $y_n = c_n(x_n - \bar{\theta})$ and $S_n = \sum_{j \geq n} y_j$. Then $S_o = 0$ and $S_n \leq 0$ for $n \neq o$. Since $y_n = S_n - \sum_{v: \text{pa}(v)=n} S_v$, finite summation by parts gives

$$\nabla W(\bar{\theta} \mathbf{1})^\top (x - \bar{\theta} \mathbf{1}) = \sum_{n \neq o} (k_n - k_{\text{pa}(n)}) S_n. \quad (38)$$

Condition (37) makes this nonpositive for every feasible x ; concavity proves global optimality. If an edge violates it, the displayed two-coordinate transfer has $S_v = -s$ and all other nonroot subtree sums zero. It therefore satisfies every continuation inequality, not just an ex ante budget. Its derivative is $\Delta > 0$, and the box gives $s_{\max}$. The curvature bound proves the finite gain. This also proves necessity without appealing to existence of unknown optimizer multipliers. $\square$

The direction moves expected payment from a descendant review to its parent. This timing respects a customer's later option to leave. By contrast, shifting payment from a low-price state to a high-price sibling can violate that sibling's continuation promise, even when total payment is unchanged. The edge criterion exposes precisely the order imposed by this institution. For weak root acceptance instead of equality, the same calculation adds $k_o S_o$; the complete scalar criterion is (37) together with $k_o \geq 0$.

A primitive curvature bound is obtained by summing all affected terms:

$$L_d \leq \sum_n w_n \bar{M}_n d_n^2 + \sum_n w_n \bar{A}_n (d_n - d_{\text{pa}(n)})^2 + \bar{G}_d, \quad d_{\text{pa}(o)} = 0. \quad (39)$$

Here $\bar{M}_n, \bar{A}_n$ bound second derivatives along the entire feasible segment, and $\bar{G}_d$ bounds the joint coupling curvature. The sum includes incoming and outgoing switching edges, including unchanged children of either

perturbed node. If $L_d = L_0 + \lambda L_A$, an untruncated step yields a guaranteed gain $\Delta^2/[2(L_0 + \lambda L_A)]$. Finite smooth adjustment can make the gain small without necessarily eliminating it.

A.2 An exact threshold for nonsmooth switching

The contrast with absolute adjustment is substantive rather than notational. Consider a two-review chain $o \rightarrow v$, $q_0 = \bar{\theta}$, the same root payment equality and child participation constraint, and switching cost $\lambda\{\kappa_o|x_o - q_0| + \kappa_v|x_v - x_o|\}$, where $\kappa_o, \kappa_v > 0$ include their probability weights. Let k_o, k_v now exclude that absolute switching term, and let the remaining resource cost be differentiable and convex. Set

$$K = \kappa_o/c_o + \kappa_v(1/c_o + 1/c_v), \quad \lambda_c = \max\{0, (k_o - k_v)/K\}.$$

Corollary A.2 (Friction threshold with an exit option). *The static protocol is globally optimal exactly when $\lambda \geq \lambda_c$. For $0 \leq \lambda < \lambda_c$, a sufficiently small customer-neutral transfer from the child to its parent strictly improves reward while satisfying continuation participation.*

Proof. Every feasible displacement has $d_o = s/c_o$, $d_v = -s/c_v$ for $s \geq 0$. Its right derivative at zero is $k_o - k_v - \lambda K$. A positive derivative gives improvement. A nonpositive derivative, combined with concavity on this one-dimensional feasible interval, rules out improvement everywhere, including the equality case. The threshold therefore uses primitive rates and frictions, not an uncomputed optimizer. $\square$

A.3 What the value of contractual memory does and does not mean

Proposition A.3 (Zero-friction bound without history-specific promises). *With exogenous regimes, compact convex tiers in $[0, 1]^d$, convex stage resources, and only ex ante linear payment commitments, assume that stage resource functions and payment coefficients depend only on time and the current regime, with no other history-dependent feasibility or physical carry-over. Let $W_{\text{full}}(\lambda)$ and $W_{t,z}(\lambda)$ allow full observed history or only time and current regime. Suppose the sole inherited-tier term is $\lambda \mathbb{E} \sum_t \beta^t \|x_t - x_{t-1}\|^2$. For any time-regime optimizer π_0 at zero friction,*

$$0 \leq W_{\text{full}}(\lambda) - W_{t,z}(\lambda) \leq \lambda \mathcal{A}(\pi_0) \leq \lambda d \sum_t \beta^t.$$

Proof. At zero friction, conditional expectation of actions given (t, z_t) preserves expected linear commitments and reduces convex stage costs. Thus both zero-friction values coincide and compactness gives π_0 . Adding nonnegative friction reduces the full value, whereas using π_0 loses at most $\lambda \mathcal{A}(\pi_0)$. Compact tier bounds give the last inequality. $\square$

The proposition deliberately excludes history-specific promises. This exclusion has content: with two equiprobable histories leading to the same terminal regime, terminal utility $x - x^2$, and respective tier caps $1/4$ and $3/4$, history-dependent control earns $(3/16 + 1/4)/2 = 7/32$, whereas a common terminal tier earns $3/16$. The gap is $1/32$ even at zero friction. Both histories and their caps are public. Different earlier installed tiers can record which promise applies. The missing information concerns a contractual obligation,

not a new signal about demand. Consequently no unconditional claim that continuation-constrained memory value vanishes is made.

For the continuation institution, a remaining-payment budget supplies the appropriate dynamic state. At node n , with incoming budget b and outside-protocol cap $\bar{b}_n$, choose the tier x and child budgets b_v subject to

$$a_n^\top x + \beta \sum_{v \in \text{ch}(n)} P_{nv} b_v \leq \min\{b, \bar{b}_n\}. \quad (40)$$

The Bellman state is (n, q, b) , or (t, z, q, b) when the subtree primitives and outside caps are Markov. Its continuation term is $\beta \sum_v P_{nv} V_v(x, b_v)$. Backward induction establishes equivalence with the full tree program: feasible child budgets give a feasible subtree protocol, while a feasible protocol supplies child budgets equal to its actual expected remaining payments. At zero friction the recursion does not use q , but it still uses the promise budget and the information determining the outside caps. This is the precise continuation-aware replacement for conditioning on demand regime alone.

Data, software, and computation accessibility

The versioned review package contains ordinary manuscript inputs, complete proofs, frozen training coefficients, generated test primitives, rational policies and multipliers, independent certificate replay, raw execution times, and source hashes. The README identifies the current build and validation entry points. Predecessor source files and scientific observations are preserved, with an explicit relocation map. All experiments are synthetic; no confidential or personal data are used.

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Table 1: Accepted continuous-tier information values

Information	Provider reward	Increment	Payment price
Static	-5.870110973	—	0.000000000
Time	-5.869294096	0.000816877	0.001568294
Time–regime	-5.252736080	0.616558016	0.174849751
Full state	-5.233445701	0.019290379	0.180271244

All four classes have the same filled demand 18.021386714, customer utility 28.657506145, and physical cost 9.587377732. All policies are deterministic. Decimals round exact rational values; stock is optimized over the full interval through Lemma 5.1.

Table 2: Certified refinement under noncentered forcing

Family	Pipeline	Met tolerance	Maximum final gap	Mean final gain	Time at 10^{-5}	Time at 10^{-7}
Cycle	Cold SLSQP	12/12	1.4e-08	0.004605	11.95	13.69
Cycle	Value	48/48	6.2e-08	0.004605	19.32	19.60
Cycle	Value–gradient	48/48	7.8e-08	0.004605	19.15	19.41
Random-edge	Cold SLSQP	12/12	8.3e-08	0.005110	30.90	35.78
Random-edge	Value	48/48	7.5e-08	0.005110	42.93	49.53
Random-edge	Value–gradient	48/48	8.8e-08	0.005110	43.30	49.91
Geometric	Cold SLSQP	12/12	9.4e-08	0.003731	54.24	67.84
Geometric	Value	48/48	9.7e-08	0.003731	73.06	88.62
Geometric	Value–gradient	48/48	9.7e-08	0.003731	73.54	86.91
Weighted block	Cold SLSQP	12/12	8.8e-08	0.001923	83.48	113.03
Weighted block	Value	48/48	8.8e-08	0.001923	110.87	140.69
Weighted block	Value–gradient	48/48	8.8e-08	0.001923	110.68	139.06

Gaps and gains are per discounted service–review. Times are median milliseconds across the declared instances and, for learned pipelines, frozen training seeds. They include candidate generation, projection, rational audit, gating, and actual refinement; a cold pipeline starts at the same optimized static protocol. Common setup is recorded separately. These are not independent timing repetitions or a speedup claim. The last column compares the same certified error tolerance, not different solver stopping flags.